
How the Two Models Connect

The economic channels of the OG-Core $\times$ CLEWS soft link —
and how each is represented in the model

numbers: the Philippine reference run · `ogclews-link` · July 6, 2026



The coupling problem

Two models, complementary blind spots

CLEWS / OSeMOSYS

a least-cost optimiser: chooses what to build and run so that energy demand is met at the lowest discounted system cost

- **rich:** technology, season-by-season operation, emissions, land & water
- **blind:** demand is *fixed* — no price response, no households, no government budget, no distribution

Each is detailed exactly where the other is blank — so what usefully crosses the seam differs by direction:

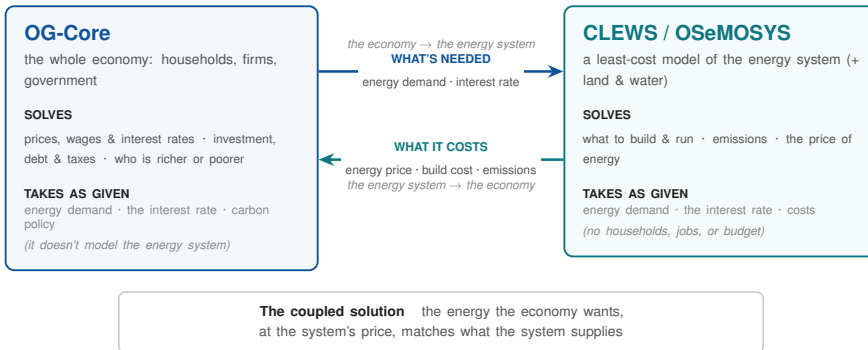
quantities flow from the economy, **prices and physical outcomes** flow from the energy system.

OG-Core

overlapping-generations general equilibrium: households by age $S \times$ lifetime income J , M industries, a government that closes its budget each period

- **rich:** behaviour, saving, fiscal policy, incidence, demography
- **blind:** no energy technology — energy is not a priced input to production

The structural seam



Each model is hollow exactly where the other is deep.

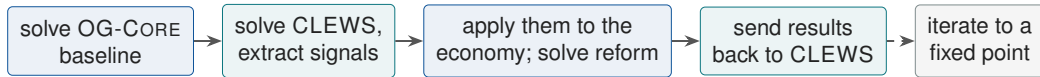
Quantities flow right (energy demand, the interest rate); prices and physical outcomes flow left (the energy price, build cost, emissions). The models stay separate and pass a few auditable signals.

Why a *staged soft* link, not one big model

A hard link is ruled out structurally, not practically:

- OG-CORE steps through time, period by period; the energy model plans all years at once
- neither fits inside the other — OG-CORE has no single objective the energy model could be nested in — and no shock can be injected mid-solve

So the models stay separate and the link runs in stages:



Validated today: the single pass (solid). The loop (dashed) is designed — the return signals already exist — but not yet wired.

Signals, wedges, and the price that carries the load

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Wedge the OG-CORE parameter that carries it: the consumption tax τ^c , an industry's productivity Z_m , public investment α_I , mortality $\rho(s, t)$, effective labour $e(t, s, j)$. (A “wedge” in the tax sense — the simplest channel inserts one between producer and consumer price.)

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Channel one signal mapped to one wedge, with a guard that fires when the mapping is large or ill-posed.

Each channel is one small, testable mapping — and nothing country-specific is built in: which industry is electricity, and how many industries there are, is read from the country's own model at run time.

The signal that links the two models: the energy price

Of everything the energy model produces, **one number** does the real work across the seam — the **price of energy**. (The model sets it by balancing supply and demand for every fuel in every season; what one more unit would cost to supply *is* that price.) It is the right hand-off because it does two things the energy model, on its own, cannot:

1. It lets demand respond

Inside the energy model, demand is a *fixed* input. Hand the price to OG-CORE households — whose energy use responds to price, down to a subsistence floor — and demand finally reacts.

2. It shows who pays

A single “system cost” is silent on who bears it. OG-CORE’s households, by age and income, split the burden — and because energy is a partial necessity, a price rise falls hardest on the poor.

So the energy price is the hand-off: the economy reacts to it, and reveals who it hits. (The two models fully *agree* once that price also equals the cost of supplying what the economy then demands — what closing the loop, later, drives toward.)

Getting that price into the economy — three routes

We have the signal — the energy price. But there is a catch: in OG-CORE, firms produce with only **capital and labour**; **energy is not a line in their costs**. So a change in the energy price has no direct way in, and we must *choose* how to represent it. Three routes, in rising fidelity:

- A tax the energy households buy** — raise the price households pay by the reform/baseline ratio. Reuses existing structure and shows incidence; but it touches only household energy, and its “revenue” is phantom. Empirically $\approx$ output-neutral.
- B raise firms’ costs** (a cost-push) — lift each industry’s cost by the energy it uses, directly and through what it buys from others, as a productivity cut. Right sign and materially contractionary; but firms cannot substitute away, so the size is only a proxy.
- C make energy a real input firms buy** — alongside capital and labour, so the price feeds their costs directly and the loop closes on quantities too. The faithful version — but it needs an extension of OG-CORE (unbuilt).

Which route you pick changes the answer: A and B carry the *same* price to opposite verdicts — roughly neutral vs. sharply contractionary. C is the destination, not a refinement. (Ahead: the energy-price channel uses A; the coupled run adds B.)



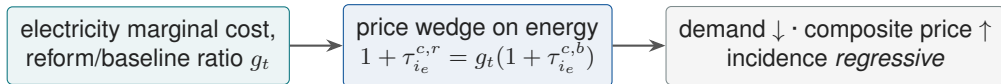
The eight channels

The map

Channel	Dir.	CLEWS signal	OG-CORE object	First-order effect
Energy price	c→o	electricity marginal cost (reform/baseline)	price wedge (τ^c) on the energy good	demand ↓; regressive incidence
Carbon price	pol	carbon price (\$/tCO ₂)	energy-wedge add-on + emissions penalty	demand ↓; emissions ↓
Public investment	c→o	grid & transmission build cost	public-investment share → public capital	public capital ↑; crowding-out, debt
Capital intensity	c→o	power-fleet capital-cost share	energy industry's capital share (γ_m)	energy price ↓; energy K ↓
Investment incentive	pol	tax-credit rate (policy response)	cost of capital (investment tax credit)	capital reallocates into energy
Health	c→o	PM _{2.5} emissions change	survival rates + worker productivity	mortality ≈ 0 ; morbidity ↑ labour
Discount rate	o→c	—	economy's return → planning discount rate	shifts the build mix (via the loop)
Energy demand	o→c	—	economy's activity → energy demand	more capacity (via the loop)

four CLEWS→OG-CORE · two policy levers applied once to both sides · two OG-CORE→CLEWS

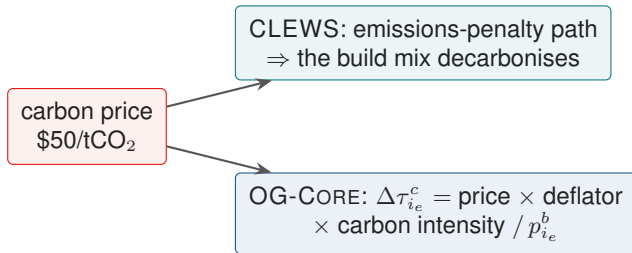
Energy price — the system's marginal cost reaches household budgets



- households face $(1 + \tau_{i_e}^c) p_{i_e}$, so energy demand responds — the price response CLEWS lacks
- poorer households spend more of their budget on energy, so they bear more of the change — even when the revenue is returned (recycled variant: $C_{ss} = -0.087\%$)
- the price is the scenario's *real* electricity price, which for the Philippines is **near-flat** — so the standalone effect is tiny ($Y_{ss} = -0.0005\%$): transmitted honestly, the data does little
- the earlier $\pm 20\%$ runs only stress-tested the *transmission*

maturity: route A, on the real price; the coupled run transmits the same price through the composite (later).

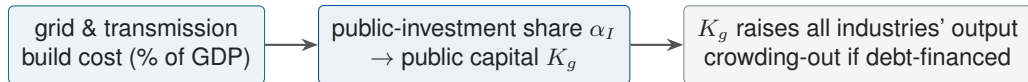
Carbon price — one price, two sides, counted once



- one chosen price enters *both* models along the same path — never read back a second time from the energy model's own implied carbon cost
- OG-CORE firms buy no energy, so only *household* energy carbon is priced on the macro side; industrial carbon is priced in CLEWS
- revenue is recyclable (transfers); standalone it is the clearest contraction: $Y_{ss} = -0.040\%$

the *level* rides on the uncalibrated unit/deflator bridge — mechanics validated, magnitude illustrative.

Public investment — the grid on the government's books



- this is where crowding-out *legitimately* lives: debt-financed spending competes for the savings pool, raises r , displaces private investment
- public-grid capital and private-generation capital touch different objects — coupling both is complementary, not double counting
- in the Philippine scenario the *extra* grid spending is ≈ 0 (the transition is generation-side), so the channel is inert here ($Y_{ss} = +0.000\%$); the mechanics scale with any non-zero spending

The capital pair I: capital intensity — a worked example

Signal: reform/baseline ratio of the power fleet's capital-cost share $\Rightarrow$ scales the energy industry's capital exponent γ_m (here +12.2%).

The model's own first-order condition (Cobb–Douglas, $\varepsilon = 1$): the user cost ρ_m and taxes cancel in the reform/baseline ratio, leaving an exact identity —

$$K_m = \frac{\gamma_m p_m Y_m}{\rho_m} \quad \Rightarrow \quad \hat{K}_E = \hat{\gamma}_E \cdot \hat{p}_E \cdot \hat{Y}_E$$

	$\hat{\gamma}_E$	$\hat{p}_E$	$\hat{Y}_E$	$\hat{K}_E$ (obs.)	identity
energy industry ($m_e = 2$)	+12.2%	−42.8%	+20.9%	−22.4%	−22.4%

- the higher exponent cuts the sector's unit cost, and competition passes that into price $\rightarrow$ electricity **collapses** −42.8%
- cheaper power lifts output +20.9% — but revenue $p_m Y_m$ still shrinks, so capital **falls** −22.4% (the identity closes to 0.03%)

Raising the capital *share* sheds capital: this is a factor-share and energy-price lever, *not* a capital draw-in.

The capital pair II: the investment incentive — the actual draw-in lever

The capital-*attraction* force is a structurally different instrument: an **investment tax credit** on the energy industry.

$$\rho_m = \frac{r + \delta - \tau_m^b \delta^\tau - \tau_m^{\text{inv}} \delta}{1 - \tau_m^b}$$

the ITC enters the *user-cost denominator*, where γ_m is absent $\Rightarrow$ capital demand shifts *out* at given r

the capital share γ_m (capital intensity)

moves factor *demand* through the production exponent; absent from ρ_m . Result: cheaper electricity, *less* energy capital.

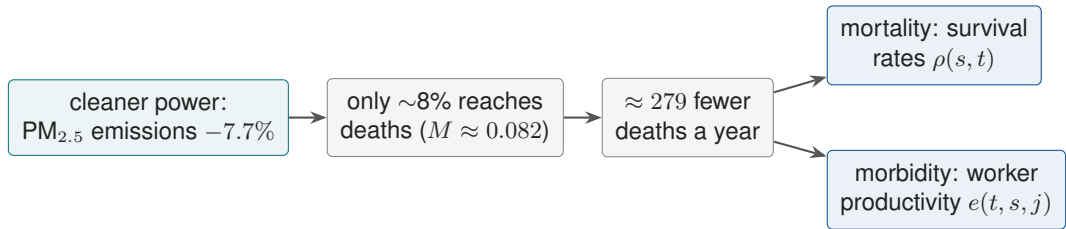
the tax credit (investment incentive)

moves the *cost of capital*; absent from the exponents. Result: capital reallocates *into* energy. (A defined lever; not part of the Philippine reference run.)

The lesson

Two levers, two different objects — *not* two views of one build-out. Intuition says “capital-intensive $\Rightarrow$ pulls capital in”; the model’s own first-order condition shows the opposite. Don’t stack them.

Health — cleaner air, fewer deaths, more productive workers



- real inputs: the Global Burden of Disease counts **43,951** deaths a year in the Philippines from breathing fine particulates (PM_{2.5}), with real age profiles
- cutting power-sector PM_{2.5} by 7.7% is *not* 7.7% fewer deaths: energy is one source among many, and deaths fall less than one-for-one as the air clears — so only ~**8%** of the cut reaches mortality ($M \approx 0.082$). That is **≈279 fewer deaths**, not the ≈3,400 from blaming energy for *every* pollution death
- **mortality**: pollution deaths fall mostly among the elderly, so the lives saved are mostly retirees — long-run output barely moves

- **morbidity**: fewer working-age illnesses means more productive workers — and *that* is the real

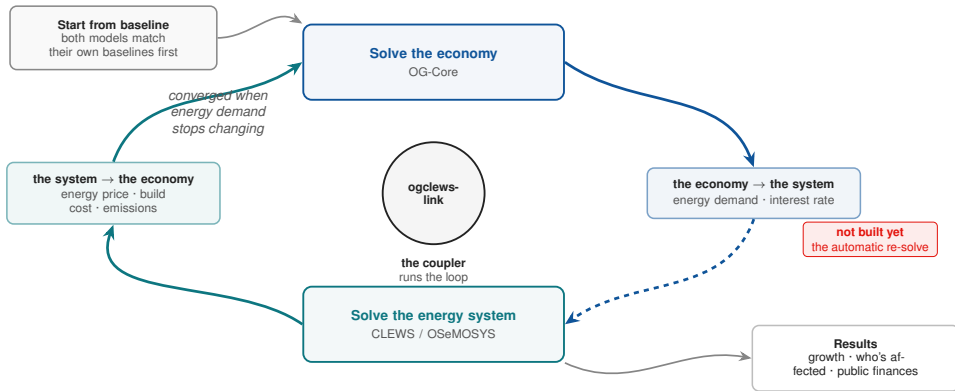
What the economy sends back: its return, its demand

Two channels run *after* the economy solves and write ready-to-use CLEWS inputs:

- **discount rate:** the economy's equilibrium return r_p becomes the energy system's planning discount rate — so energy planning uses the *model-consistent* cost of capital. A lower return favours capital-heavy, long-lived assets.
- **energy demand:** how activity changed (industry output or consumption, with an elasticity) scales the demand the system must meet. In a single pass the change is $\approx$ zero — latent by construction.

They move the macro economy only once the loop iterates — the quantity link of the classic coupled energy–macro models (MESSAGE-MACRO, TIMES-MACRO), but feeding an economy with households and a government instead of a single representative agent.

Closing the loop



Re-solve the two models in turn until energy demand stops changing. Validated today: the single pass; naive price feedback can oscillate, so the loop needs damping.



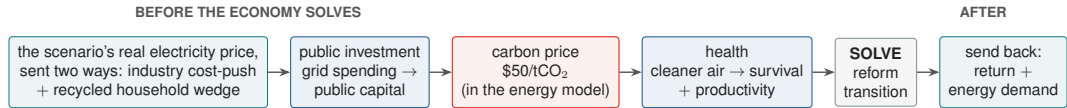
Counting each signal once

Five principles, each enforced by a guard

1. **One carbon price.** It enters once — a CLEWS penalty and an OG-CORE wedge on the same path — never *also* read back from the energy model's implied carbon cost.
2. **One view of an energy-cost shock.** A price wedge, a carbon wedge, and a productivity haircut are alternative doors for the *same* cost — never stacked.
3. **One view of generation capital.** A cost-push *or* an investment credit — and the capital share (γ_m) is *not* a third interchangeable view: it is a factor-income lever that sheds capital (the worked example).
4. **Revenue must be routed.** A τ^c wedge mechanically raises consumption-tax revenue; recycle it (transfers, public investment) or finance it explicitly — else a hidden fiscal expansion rides along.
5. **Public and private capital are distinct.** $\alpha_I \rightarrow K_g$ and the firm's own K touch different objects; coupling both is complementary.

Does it behave? The Philippine coupled run

Anatomy of the flagship: the coupled run

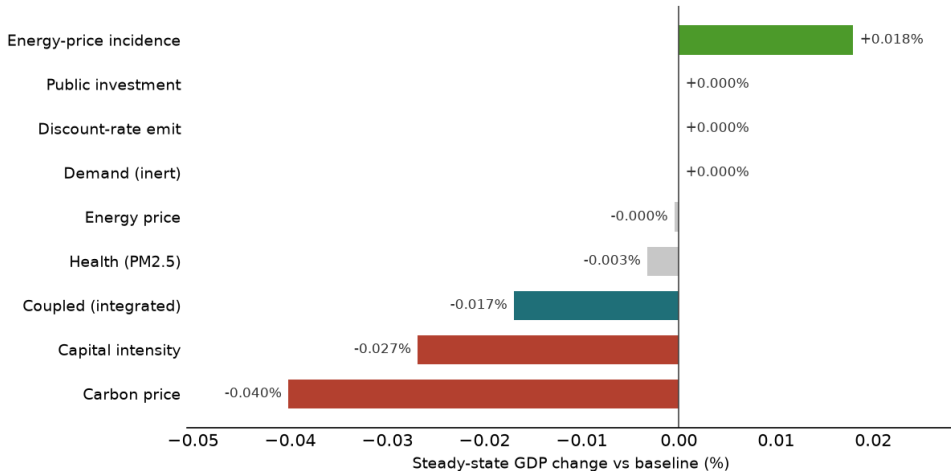


- the Philippine electricity price is near-flat $\Rightarrow$ the energy channels contribute almost nothing (electricity output moves -0.18%); **the macro response is carried by health**
- here the carbon price acts in the energy model (one-price principle); the -0.040% shown earlier is the standalone household-wedge version

% change from baseline	steady state	t_0	t_{10}
output Y	-0.017	$+0.165$	$+0.395$
consumption C	$+0.008$	$+0.195$	$+0.467$

PHL OG×CLEWS — steady-state macro impact by channel

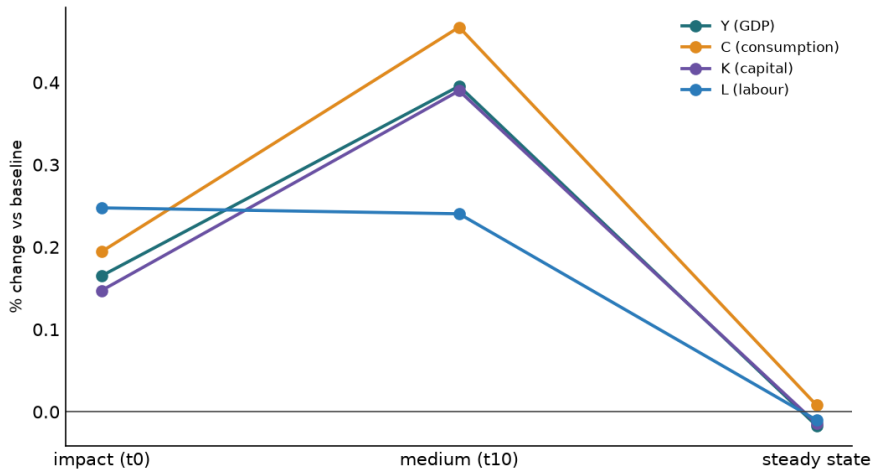
M=8 · ogcore 0.16.3 · real CLEWS 'auto' electricity price (no illustrative +20%)



Per-channel steady-state output impact, each channel run alone on the Philippine calibration. The energy price is ≈ 0 on the near-flat real price; the carbon price (-0.040%) is the clearest contraction; capital intensity is -0.027% in aggregate (but reshapes the energy sector); health is -0.003% at steady state yet $+0.19\%$ near-term.

PHL coupled result over the transition

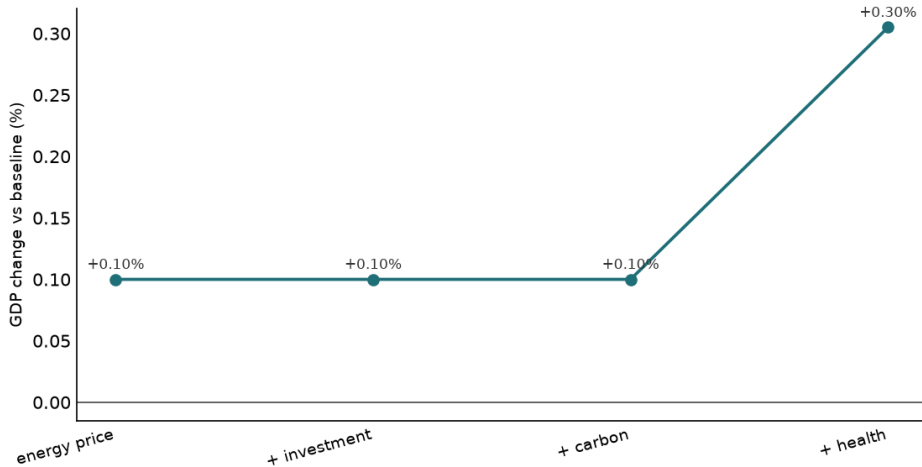
real CLEWS energy + public investment + carbon + GBD health · M=8 · ogcore 0.16.3



The coupled transition: steady state -0.017% , but $+0.165\%$ at t_0 rising to $+0.395\%$ by t_{10} — the working-age morbidity gain along the path, added retirees in the long run. The horizon split is the health channel's signature, and it is invisible to a static model.

Cumulative channel contributions to the real coupled result

each step adds one channel; top step = coupled · M=8 · real CLEWS energy



Cumulative decomposition: channels layered so the top layer *equals* the coupled run — every effect attributable to a named mechanism. That auditability is the point of the channel architecture.

Status

Where the coupling stands

Validated today

- eight channels, each one signal → one wedge, guard-railed
- single CLEWS→OG-CORE pass on *real* inputs: the scenario's real electricity price, the real Global Burden of Disease data
- each model runs in its own software environment; the link passes a few auditable files between them
- every number locked in a reference record and re-checked by tests; signs and mechanics as theory predicts

Open, in order

- **loop closure** — the emitters are ready; the outer re-solve-and-iterate is not wired
- **the unit/deflator bridge** — carbon and investment *levels* are illustrative until it is calibrated
- **Route C** — energy as a priced production input: the structural destination

One line: A small, auditable library of economic mechanisms — each signal counted once, each effect attributable — rather than one opaque coupling.